Smoke Detector Alarm System using MQ135 Sensor and Arduino UNO

Reason Behind

Smoke and harmful gases are major causes of fire hazards and health-related issues in homes, laboratories, and industries. This is why I designed a simple smoke detection system using an MQ135 gas sensor and Arduino UNO. It can detect harmful gases and provide instant alerts to user by using LCD screen, indicating LEDs & a Buzzer.

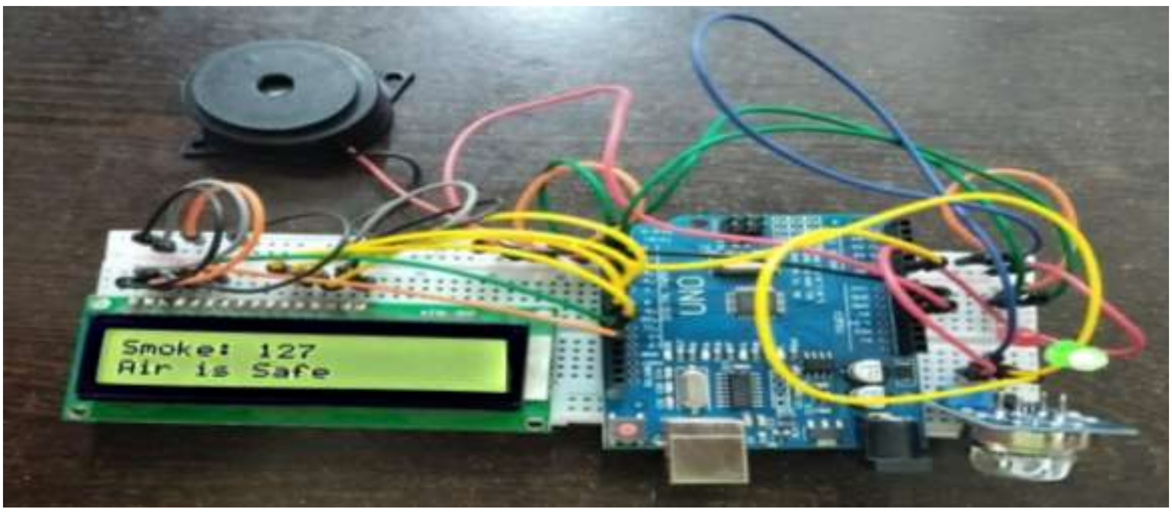
Although it's a simple project and can be found online very easily, but my motto to make this project is to understand how sensors, microcontrollers, displays, and output devices work together to create a complete embedded system.

Components Required

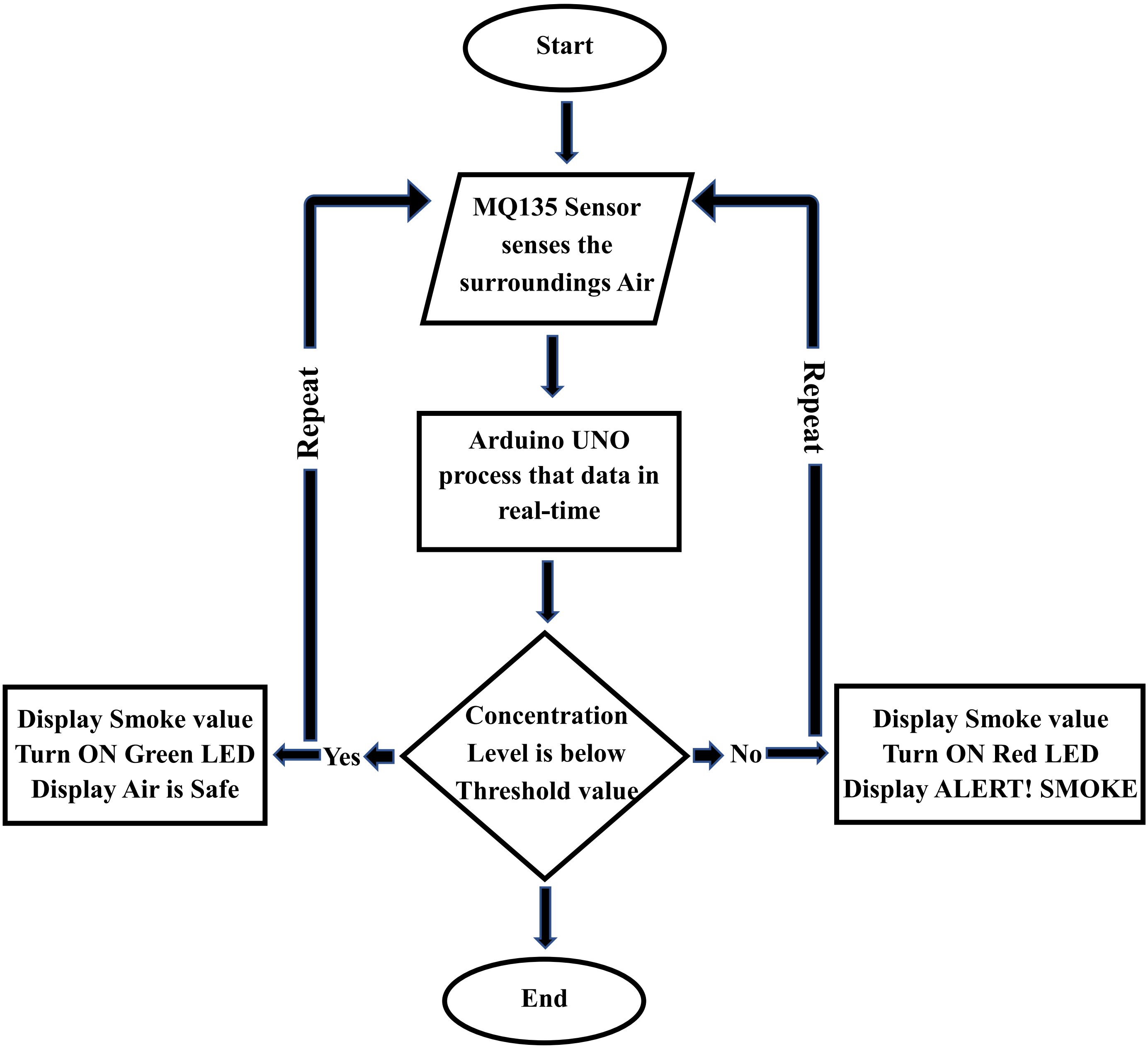
- Arduino UNO
- MQ135 Gas Sensor
- 16×2 LCD
- Red LED
- Green LED
- Active Buzzer
- 220Ω Resistors
- Breadboard
- Jumper Wires
- USB Cable

Working Principle

It continuously monitors the surrounding air using an MQ135 gas/smoke sensor. In normal conditions, a green LED glows with a message on LCD that, the Air is safe and there'll be no beep sound. But when smoke or harmful gases exceed the pre-set threshold, the Arduino sends the signal to high the red LED pin, activates the buzzer, and displays a warning message on a 16×2 LCD.



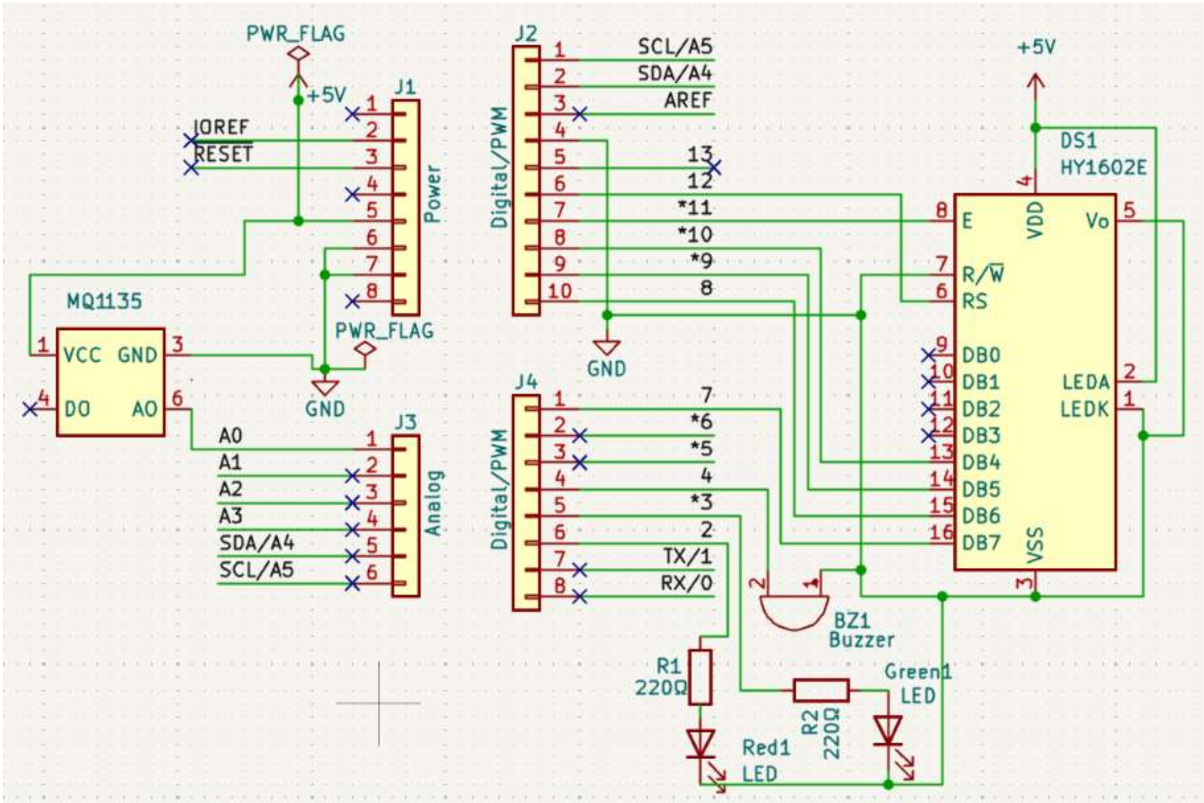
Working Process Flow



Circuit Description

Components	Arduino Board Pin
MQ 135 Sensor	A0
16x2 LCD	12,11,10,9,8,7
Red LED	2
Green LED	3
Buzzer	4

Full Schematic



Advantages

- Low Cost & Easy to Build
- Real-time monitoring
- Portable system
- Provides instant alerts

Limitations

- Requires Calibration
- Sensitive to Multiple Gases
- Not Industrial Grade
- Humidity affects readings

Conclusion

With the help of this project, we can measure the pollution level in our surroundings so that, we may advice people to not stay longer in that environment & also, we can initiate the steps to reduce the toxicity level from the atmosphere.